

Thus, rearranging Eqs. (7.5.17), (7.5.18), and (7.5.22), we obtain two separate systems of six equations in six unknowns:

the **linearized longitudinal equations**,

$$\begin{bmatrix} W/g - F_{x_b, \dot{u}} & -F_{x_b, \dot{w}} & 0 & 0 & 0 & 0 \\ -F_{z_b, \dot{u}} & W/g - F_{z_b, \dot{w}} & 0 & 0 & 0 & 0 \\ -M_{y_b, \dot{u}} & -M_{y_b, \dot{w}} & I_{yy_b} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \Delta \dot{u} \\ \Delta \dot{w} \\ \Delta \dot{q} \\ \Delta \dot{x}_f \\ \Delta \dot{z}_f \\ \Delta \dot{\theta} \end{Bmatrix} = \begin{Bmatrix} F_{x_b, \delta_e} \\ F_{z_b, \delta_e} \\ M_{y_b, \delta_e} \\ 0 \\ 0 \\ 0 \end{Bmatrix} \Delta \delta_e \quad (7.5.23)$$

$$+ \begin{bmatrix} F_{x_b, u} & F_{x_b, w} & F_{x_b, q} & 0 & 0 & -W \cos \theta_o \\ F_{z_b, u} & F_{z_b, w} & F_{z_b, q} + V_o W/g & 0 & 0 & -W \sin \theta_o \\ M_{y_b, u} & M_{y_b, w} & M_{y_b, q} & 0 & 0 & 0 \\ \cos \theta_o & \sin \theta_o & 0 & 0 & 0 & -V_o \sin \theta_o \\ -\sin \theta_o & \cos \theta_o & 0 & 0 & 0 & -V_o \cos \theta_o \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta x_f \\ \Delta z_f \\ \Delta \theta \end{Bmatrix}$$

and the **linearized lateral equations**,

$$\begin{bmatrix} W/g & 0 & 0 & 0 & 0 & 0 \\ 0 & I_{xx_b} & -I_{xz_b} & 0 & 0 & 0 \\ 0 & -I_{xz_b} & I_{zz_b} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \Delta \dot{v} \\ \Delta \dot{p} \\ \Delta \dot{r} \\ \Delta \dot{y}_f \\ \Delta \dot{\phi} \\ \Delta \dot{\psi} \end{Bmatrix} = \begin{Bmatrix} F_{y_b, \delta_a} & F_{y_b, \delta_r} \\ M_{x_b, \delta_a} & M_{x_b, \delta_r} \\ M_{z_b, \delta_a} & M_{z_b, \delta_r} \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{Bmatrix} \begin{Bmatrix} \Delta \delta_a \\ \Delta \delta_r \end{Bmatrix} \quad (7.5.24)$$

$$+ \begin{bmatrix} F_{y_b, v} & F_{y_b, p} & F_{y_b, r} - V_o W/g & 0 & W \cos \theta_o & 0 \\ M_{x_b, v} & M_{x_b, p} & M_{x_b, r} & 0 & 0 & 0 \\ M_{z_b, v} & M_{z_b, p} & M_{z_b, r} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & V_o \cos \theta_o \\ 0 & 1 & \tan \theta_o & 0 & 0 & 0 \\ 0 & 0 & \sec \theta_o & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta y_f \\ \Delta \phi \\ \Delta \psi \end{Bmatrix}$$

Equations (7.5.23) and (7.5.24) are the linearized 6-DOF equations of motion. They provide an important simplification over the more general nonlinear 6-DOF equations in Sec. 7.4. However, we must remember that these equations are valid only if the motion of the aircraft is confined to small deviations about some steady equilibrium flight condition. Larger deviations will introduce coupling between the lateral and longitudinal motions and nonlinearity in the aerodynamic forces and moments.